A conservative receiver and the origin of a correlation scale

A finite harmonic receiver has a vanishing long-window action observable at fixed centre velocity. An increasing periodic chain instead has plateau $\Theta\sqrt{m/k}$ when its size tends to infinity first, with mode energy Θ and spring stiffness k. A common hard particle-speed ceiling on the stated phase preparation forces this plateau to close. The comparison locates the positive scale in the low-frequency weights and energy preparation.

1. Hamiltonian and invariant preparation

Let $x, p \in \mathbb{R}^n$, $n \geq 2$, $M = \operatorname{diag}(m_1, \ldots, m_n)$ with $m_i > 0$, and let K be a connected spring-network Laplacian. Thus $K = K^T \geq 0$ and $\ker K = \operatorname{span}\{\mathbf{1}\}$. The Hamiltonian and equations are

$$\mathcal{H} = \tfrac{1}{2} p^T M^{-1} p + \tfrac{1}{2} x^T K x, \qquad \dot{x} = M^{-1} p, \quad \dot{p} = -Kx.$$

Energy and total momentum are conserved. These linear equations give global smooth trajectories. Write $D = M^{-1/2} K M^{-1/2}$ and choose an orthonormal eigenbasis $e_0, e_1, \ldots, e_{n-1}$, with $e_0 = M^{1/2}\mathbf{1}/\sqrt{M_{\text{tot}}}$ and positive eigenvalues ω_j^2 for $j \geq 1$. Define canonical coordinates $q_j = e_j^T M^{1/2} x$, $P_j = e_j^T M^{-1/2} p$.

Fix centre velocity $V_0 = 0$ and centre position. Assign internal mode energies $E_j \geq 0$, and independent phases ϕ_j uniform on $[0, 2\pi)$. Then

$$q_j(t) = \frac{\sqrt{2E_j}}{\omega_j} \cos(\omega_j t + \phi_j), \qquad P_j(t) = -\sqrt{2E_j} \sin(\omega_j t + \phi_j).$$

This is an invariant probability ensemble on the internal phase tori, even when frequencies are commensurate. It is a preparation, not a mixing premise. The total energy is $E = \sum_j E_j$, so $|\dot{x}_i| \leq \sqrt{2E/m_i}$ for every trajectory. Choosing $E < \frac{1}{2}(\min_i m_i)c^2$ puts all particles below c in this frame. These are Newtonian springs with a speed-bounded preparation; propagation of signals and Lorentz covariance are separate model requirements.

2. Exact action observable and its limits (C047)

For tagged particle i, let $b_{ij} = e_{ji}/\sqrt{m_i}$, where e_{ji} denotes component i of vector e_j. Then $v_i(t) = \sum_{j \geq 1} b_{ij} P_j(t)$ has zero mean and covariance

$$C_i(t) = \sum_{j \geq 1} w_{ij} \cos(\omega_j t), \qquad w_{ij} = b_{ij}^2 E_j \geq 0.$$

Phase averaging gives this identity: distinct modes have zero cross covariance, and one phase has $\mathbb{E}[P_j(t)P_j(0)] = E_j \cos(\omega_j t)$. Integrating the covariance over the displacement square yields, for $\Delta > 0$,

$$\mathsf{h}_i(\Delta) = \frac{m_i}{\Delta} \operatorname{Var}[x_i(t+\Delta) - x_i(t)] = \frac{2m_i}{\Delta} \sum_{j \geq 1} \frac{w_{ij}}{\omega_j^2} [1 - \cos(\omega_j \Delta)].$$

The observable has action units and is independent of preparation time t. Set $B_i = \sum_j w_{ij}/\omega_j^2$, with units of length squared. Then

$$0 \leq \mathsf{h}_i(\Delta) \leq \frac{4 m_i B_i}{\Delta} \longrightarrow 0 \quad (\Delta \to \infty).$$

At short windows, $\mathsf{h}_i(\Delta) = m_i C_i(0)\Delta + O(\Delta^3)$. The finite cosine correlation has persistent memory; its ordinary improper Green–Kubo integral need not exist. The finite-window formula above supplies the relevant limit directly, without replacing the Hamiltonian dynamics by a reversible dissipative generator.

3. Centre motion and the two-body receiver (C047)

An independent random constant centre velocity with variance σ_0^2 adds $m_i\sigma_0^2\Delta$ to h_i. This follows from $x_i(t+\Delta)-x_i(t) = V_0\Delta+$ internal displacement. The velocity ensemble is stationary; for nonzero centre motion the absolute-position ensemble need not be. A deterministic centre velocity contributes zero to the variance and can be removed by a change of frame.

For two particles of masses m, M_r coupled by a spring $k > 0$, put $\mu = mM_r/(m + M_r)$ and $\omega^2 = k/\mu$. At zero total momentum, $x_1 = R + [M_r/(m + M_r)]r$, and assign relative energy E with uniform phase. The tagged response is

$$\mathsf{h}_1(\Delta) = \frac{2mE}{k\Delta}\left(\frac{M_r}{m + M_r}\right)^2 [1 - \cos(\omega\Delta)].$$

The spring is a genuine momentum receiver: $m\ddot{x}_1 = -kr$ and $M_r\ddot{x}_2 = kr$. Its reversible exchange produces the oscillatory response rather than the exponential memory law supplied by A02's refreshed bath. Scaling every E_j by a^2, $0 < a \leq 1$, scales the entire response by a^2 while preserving the equations, conservation laws and speed margin.

4. What the large-receiver test must change (C047)

For a sequence of receivers and windows $\Delta_N \to \infty$, the bound $\mathsf{h}_{i,N}(\Delta_N) \leq 4m_{i,N}B_{i,N}/\Delta_N$ shows that a positive limiting response requires this upper bound to stay positive. In particular, uniform control of $m_{i,N}B_{i,N}$ forces a zero limit. Soft modes can enlarge $B_{i,N}$, but their weights depend on the measured coordinate and the preparation. This is the next selection test.

The finite model yields no physical-time attraction from the invariant preparation: the observable already has no t dependence. Next specify an increasing spring network, its energy allocation and tagged spectral measure, then compare taking receiver size and observation duration to infinity in opposite orders. Derive any surviving constant from these mechanical inputs. Mesh refinement of the same smooth trajectories is a third, distinct limit.

5. An increasing periodic chain (C048)

Take odd $N \geq 3$ equal masses m on a ring, with spring energy $\frac{k}{2}\sum_{i=0}^{N-1}(x_{i+1} - x_i)^2$ (indices modulo N). Fix the centre position and momentum. Give every real internal normal mode energy $\Theta > 0$ and independent uniform phase. Here Θ is a preparation energy, and total energy is $(N - 1)\Theta$. The mode frequencies and tagged covariance are

$$\omega_j = \Omega\sin(\pi j/N), \quad \Omega = 2\sqrt{k/m}, \qquad C_N(t) = \frac{\Theta}{mN}\sum_{j=1}^{N-1}\cos(\omega_j t).$$

The real sine/cosine eigenvectors have equal mode energies, so their paired squared components sum to $2/N$ at each site. Applying §2 gives

$$h_N(\Delta) = \frac{2\Theta}{N\Delta}\sum_{j=1}^{N-1}\frac{1 - \cos(\omega_j\Delta)}{\omega_j^2}.$$

For each fixed Δ, this Riemann sum converges to

$$h_\infty(\Delta) = \frac{2m}{\Delta}\int_0^\Omega \frac{1 - \cos(\omega\Delta)}{\omega^2}\rho(\omega)\,d\omega, \quad \rho(\omega) = \frac{2\Theta}{\pi m\sqrt{\Omega^2 - \omega^2}}.$$

This is a statement about limits of tagged covariances and increment variances; it does not require a stationary absolute-position law for the infinite chain. The density is integrable at its upper edge and continuous at zero. Splitting at any fixed $0 < a < \Omega$, the contribution from $[a,\Omega]$ is $O(1/\Delta)$. In the lower interval substitute $y=\omega\Delta$. Dominated convergence, using $(1-\cos y)/y^2 \in L^1(0,\infty)$ and bounded ρ on $[0,a]$, gives

$$\lim_{\Delta\to\infty} h_\infty(\Delta) = \pi m\rho(0) = \frac{2\Theta}{\Omega} = \Theta\sqrt{m/k}.$$

The integral $\int_0^\infty (1-\cos y)y^{-2}dy = \pi/2$ follows by integration by parts and the Dirichlet sine integral. Thus

$$\lim_{\Delta\to\infty}\lim_{N\to\infty} h_N(\Delta) = 2\Theta/\Omega, \qquad \lim_{N\to\infty}\lim_{\Delta\to\infty} h_N(\Delta) = 0.$$

A joint regime is also explicit. Write $f(z) = 2(1-\cos z)/z^2 = 2\int_0^1 (1-r)\cos(zr)\,dr$ with $f(0)=1$. Then $|f'| \le 1/3$. The integrand $\Theta\Delta f(\Omega\Delta\sin\pi x)$ is Lipschitz with constant at most $\pi\Theta\Omega\Delta^2/3$. A left Riemann sum error is at most that constant divided by $2N$; removing its zero mode contributes $\Theta\Delta/N$. Therefore

$$|h_N(\Delta) - h_\infty(\Delta)| \le \frac{\pi\Theta\Omega\Delta^2}{6N} + \frac{\Theta\Delta}{N}.$$

At fixed m,k,Θ, $\Omega\Delta_N \to \infty$ and $(\Omega\Delta_N)^2/N \to 0$ give the positive joint limit $2\Theta/\Omega$. The action scale is selected by the input energy and spring timescale.

6. Energy and a common speed ceiling (C049)

The positive-plateau preparation has extensive total energy. Its phase support also contains increasingly large tagged velocities. At site zero use the standard real Fourier basis. Only the $(N-1)/2$ cosine modes contribute; each has maximal velocity contribution $2\sqrt{\Theta/(mN)}$. Phases can align, so the exact support maximum is

$$\sup_\phi |v_0| = (N-1)\sqrt{\Theta/(mN)}.$$

Every neighbourhood of the maximizing phases has positive preparation measure. A common ceiling $|v_i| \le c$ for all supported trajectories thus requires $\Theta_N \le mc^2N/(N-1)^2 = O(1/N)$. This is a necessary condition from one site; no sufficiency for all sites is inferred from that maximum.

Both bounded total energy and this necessary speed condition close the response uniformly in the window. Indeed, $1-\cos z \le z^2/2$ gives $h_N(\Delta) \le \Theta_N\Delta$. The identity $\sum_{j=1}^{N-1}\csc^2(\pi j/N) = (N^2-1)/3$ gives

$$h_N(\Delta) \le \frac{4\Theta_N(N^2-1)}{3N\Omega^2\Delta}, \qquad \sup_{\Delta>0} h_N(\Delta) \le \frac{2\Theta_N}{\Omega}\sqrt{\frac{N^2-1}{3N}}.$$

The trigonometric identity follows by differentiating $\sum_{j=0}^{N-1}\cot(x+\pi j/N) = N\cot(Nx)$ and subtracting the $j=0$ singularity as $x\to 0$. The last inequality uses $\min(A\Delta, B/\Delta) \le \sqrt{AB}$. For $\Theta_N = O(1/N)$ the uniform bound tends to zero. Thus this explicit positive reservoir plateau uses a different preparation class from a family with a common strict particle-speed ceiling. Its dependence on Θ also leaves preparation-independent action selection open.

7. Source input and reproduction

Ford–Kac–Mazur, printed p. 505, (2)–(5), supplies the matrix harmonic propagator. Their canonical preparation excludes zero eigenvalues at that stage; ours fixes the centre and uses compact fixed-energy phase tori. Zwanzig, p. 219, (21)–(24), supplies a complementary reduced-description route: eliminate harmonic bath coordinates to obtain a cosine memory kernel. That friction kernel differs from the tagged velocity covariance calculated here. Its continuum replacement must be propagated through the tracer equation before identifying a displacement coefficient.

B23 records the established ingredients, derived finite-network consequence and bounded reading coverage. Historical modal, two-body and three-body checks accompany C047's proof. C047 has coordinator proof review and a sequential Luna-low prior-art audit. Its limits concern the stated observable and preparation class.

B24 audits §§5–6 against Ford–Kac–Mazur's cyclic Fourier construction and its finite-sum/infinite-chain distinction, printed p. 506, (12)–(24). The tagged density, plateau and speed-support bounds are derived here with fixed-energy phases. Six exact identities, four finite ring spectrum/cosecant comparisons and four Riemann-bound tests were recorded before the hard no-Python-verification rule; the handoff records that history. The new uncommitted verification script and output were withdrawn under the rule. Current mathematical verification uses the written proofs and source review. The next mechanism must retain bounded velocities while supplying low-frequency response; its preparation dependence remains a separate selection test.

8. What information survives a cut?

The fixed-energy two-body receiver gives an exact cut-state countertest: position-only stationary conditional kernels fail composition, while retaining momentum restores it. Independently refreshing direction at every inserted cut preserves one-time position marginals but freezes terminal position as the mesh shrinks. These are C062–C063; the full maps and proof are in the cut-state note, with the B33 audit.

With $q = x - X$, $\mu = mM/(m + M)$, $\omega = \sqrt{k/\mu}$ and $A = \sqrt{2E/k}$, uniform phase on the energy ellipse gives the conditional one-segment positions

$$q' = q\cos\omega t \ \pm \sqrt{A^2 - q^2}\,\sin\omega t$$

with equal weights. Thus the full mean after $s + t$ is $q\cos\omega(s + t)$, whereas independent conditional composition gives $q\cos\omega s\cos\omega t$. The missing interface variable is $p_s = \mu\dot{q}_s$ in $q_{s+t} = q_s\cos\omega t + p_s\sin\omega t/(\mu\omega)$. It has momentum units. Keeping this full internal state composes exactly; marginalizing the true joint path law also preserves all cut restrictions.

For N independent resets at intervals T/N, direct iteration gives

$$\mathbb{E}Q_N = q[\cos(\omega T/N)]^N,$$
$$\mathbb{E}Q_N^2 = \frac{A^2}{2} + \left(q^2 - \frac{A^2}{2}\right)[\cos(2\omega T/N)]^N.$$

The first limit is q and the second is q^2, proving terminal conditional L^2 convergence to the starting position. Under the original stationary preparation the reset two-time correlation tends to $A^2/2$, while the true value is $(A^2/2)\cos\omega T$. A cut that only observes the motion performs no reset and leaves the true correlation unchanged.

The normalized orbital action is $J = (2\pi)^{-1}\oint p\,dq = E/\omega$: integrating $\mu\omega A^2\sin^2\phi$ over a cycle proves the formula. Uniform-phase covariance also has square-root determinant E/ω. Cooling $E \downarrow 0$ at fixed Hamiltonian closes both, with exact phase-state composition intact. R04 extends this test to a third body, as follows.

9. A third body: exact memory and recoverable history

Retaining tagged momentum leaves receiver memory in the equal-mass open three-body chain. At zero centre position and momentum, let $x = x_1$, $y = x_2 - x_3$, $\mu = 3m/2$, $\nu = m/2$, $P = \mu\dot{x}$ and $Q = \nu\dot{y}$. Physical tagged momentum is $p_1 = 2P/3$. The internal Hamiltonian is

$$H = \frac{P^2}{2\mu} + \frac{Q^2}{2\nu} + \frac{9k}{8}x^2 - \frac{3k}{4}xy + \frac{5k}{8}y^2.$$

For fixed microcanonical energy $E > 0$, eliminate y, Q exactly. With $g = 3k/4$, $d = 5k/4$ and $\Omega^2 = 5k/(2m)$,

$$\mu\ddot{x} + \frac{9k}{5}x + \int_0^t \frac{9k}{20}\cos\Omega(t-s)\dot{x}(s)\,ds = F_0(t),$$

$$F_0(t) = g\left(y_0 - \frac{g}{d}x_0\right)\cos\Omega t + \frac{gQ_0}{\nu\Omega}\sin\Omega t.$$

These inherited quadratures carry the receiver information across a cut. The kernel has stiffness units; its convolution is a force. The full proof in R04 also recovers the hidden pair from tagged phase at two sufficiently close times. The relevant block determinant is $g^2\delta^4/(12\mu\nu) + O(\delta^6)$; inverse entries grow up to δ^{-3}. Exact recovery and finite-precision physical readout therefore pose different requirements.

The internal frequencies are $\sqrt{k/m}$ and $\sqrt{3k/m}$, and modal actions are E_j/ω_j. Cooling scales the actions to zero while keeping the memory kernel fixed. C064–C065 and B34 record this derived specialization of harmonic-bath elimination and observability. R05 supplies the following readout test.

10. Readout precision and a scalable classical instrument

Two sequential classical probes read the tagged pair with vanishing error and back-reaction under increasingly concentrated preparations. With canonical probe pairs (q, π), (r, ρ) and integrated interaction Hamiltonians $G_x = \alpha x\pi$, $G_P = \beta P\rho$, the pre-cut estimates and kicks are

$$\widehat{x} = x + q/\alpha, \qquad \widehat{P} = P + r/\beta - \alpha\pi,$$
$$d_x = \beta\rho, \qquad d_P = -\alpha\pi.$$

Here α is dimensionless and β has units T/M. For incoming rectangular support half-widths (s_q, t_q, s_r, t_r), the intrinsic support accuracy-disturbance products are $s_q t_q$ and $s_r t_r$, in action units. Both can approach zero across ordinary positive-volume classical preparations.

At fixed horizon T, put $\eta = (T/N)/t_*$ with fixed time unit t_*. Widths $O(\eta^4)$ yield hidden-state recovery error $O(\eta)$ after the cubic inverse amplification, including the intervening kick. The accumulated state disturbance is $O(\eta^3)$ uniformly over the horizon. The original receiver preparation stays fixed, while fresh probe count and total fixed-mass probe mass grow as $O(\eta^{-1})$.

C066–C067, the full readout proof and B35 specify this externally switched impulsive instrument. Its ideal position records and shrinking probe widths leave preparation and recording resources open. R06 supplies the finite-duration construction below; readout conditioning and canonical reaction alone have not excluded zero action.

11. Finite-duration readout with an autonomous clock

One mechanical clock and four free probes suffice for delayed reconstruction of the three-body receiver at fixed total apparatus mass. Let (s, p_s) be the clock pair, (q_j, π_j) the probe pairs, and all five masses be positive and fixed. With fixed stiffness K and dimensionless $0 < \lambda \le \lambda_0$, the autonomous Hamiltonian is

$$H_\lambda = H + \frac{p_s^2}{2M_c} + \sum_{j=1}^{4}\frac{\pi_j^2}{2M_j} + \lambda K \sum_{j=1}^{4} f_j(s)X(x)R(q_j).$$

The functions X, R are bounded smooth coordinate cutoffs, equal to their arguments on the admitted trajectories. Fixed smooth bumps f_j switch the couplings as a prepared clock moves through them. Pulse widths and observation time $T > 0$ stay fixed. The interaction forces are globally bounded; harmonic forces are uniformly bounded on the fixed receiver energy preparation and bounded apparatus energy class. All kinetic terms are positive.

The reference output $x^0(t)$ determines $z_0 = (x_0, P_0, y_0, Q_0)$: its first four derivative rows have triangular diagonal $1, 1/\mu, g/\mu, g/(\mu\nu)$. Analyticity supplies four independent time evaluation rows. Choosing sufficiently narrow but fixed positive-width bumps preserves independence of the integrated matrix

$$\mathcal{A}_{j\cdot} = K \int_0^T f_j(s_0 + vt) e_x \Phi_t \, dt.$$

For incoming apparatus support widths b in fixed component units, Hamilton's equations and finite-horizon integration give

$$\boldsymbol{\pi}(T) = \boldsymbol{\pi}(0) - \lambda \mathcal{A} z_0 + O(\lambda b + \lambda^3), \qquad \sup_t \|z(t) - \Phi_t z_0\| = O(\lambda b + \lambda^2).$$

These estimates include clock reaction. The clock leaves the pulse supports with positive momentum, after which the four probe momenta are conserved. Exact access to them gives the calibrated estimate $\widehat{z}_0 = -\mathcal{A}^{-1}\boldsymbol{\pi}(T)/\lambda$, with error $O(b/\lambda + b + \lambda^2)$. Choosing $b = \lambda^3$ makes reconstruction and disturbance both $O(\lambda^2)$ at fixed nonzero clock mean energy. Initial canonical position error times maximum canonical momentum disturbance is then $O(L_* P_* \lambda^4)$, in action units for fixed units L_*, P_*.

C068–C069 and the full autonomous proof specify the compact preparation domain, cutoff continuation and uniform estimates; B37 audits the pointer and observability precedents. The result uses increasingly precise preparation, exact final records and known dynamics. All reported cuts may be refined from the same four records after fixed latency T. R07 supplies the following separate new-information test. The upper mass, duration and force resources alone leave the autonomous readout's action product with zero infimum.

12. A delayed observer with an unresolved force

A single-particle information task yields a sharp prediction lower bound. Keep exact records through time zero, including (q_0, p_0), but allow no new record before the prediction time $\ell > 0$. Let the unknown external force be any measurable f with $|f| \le F$. This explicitly differs from the reconstructed known dynamics above. For mass $m > 0$,

$$v := p(\ell) - p_0 = \int_0^\ell f(s) ds, \qquad u := q(\ell) - q_0 - p_0 \ell/m = \frac{1}{m} \int_0^\ell (\ell - s) f(s) ds.$$

The forces $+F$ and $-F$ have identical accessible records. Their endpoint separations imply worst-case prediction errors of at least $F\ell^2/(2m)$ and $F\ell$. Inertial prediction attains both, so the product of coordinate minimax errors is exactly $F^2\ell^3/(2m)$, in action units.

The joint region carries more information than these two error bars. Writing $d = v/(F\ell)$ and $z = mu/(F\ell^2)$, it is exactly

$$-1 \le d \le 1, \qquad |z - d/2| \le (1 - d^2)/4.$$

At fixed impulse, maximal early positive force and maximal late negative force give the upper boundary; reversing their order gives the lower one. Convex combinations fill the region. Integrating its width gives canonical area $2F^2\ell^3/(3m)$, without a 2π normalization. This area describes admissible alternatives, not a lower action for each individual motion.

C070–C071: proof, source audit. The area closes with force budget or delay. R08 supplies the cut-composition result below.

13. Exact cuts and observed-position information

At fixed total horizon $T = a + b$, an unobserved cut preserves the bounded-force reachable set exactly. Write $S_b(q, p) = (q + bp/m, p)$ and let K_t be the increment lens above with duration t. Splitting and concatenating admissible force inputs proves

$$K_T = S_b K_a + K_b.$$

The sum is Minkowski addition; every finite unobserved refinement therefore leaves the old endpoint region and its canonical area unchanged. Replacing K_a by its marginal rectangle admits incompatible position–momentum pairs and strictly enlarges the propagated set.

An exact, non-disturbing phase-state record at the cut leaves a translate of K_b, with area $2F^2b^3/(3m)$. An exact position record instead leaves a momentum interval of width

$$D = Fa\,[\sqrt{2-4\zeta} + \sqrt{2+4\zeta} - 2], \qquad \zeta = \frac{m(q_a - q_0 - p_0 a/m)}{Fa^2} \in [-1/2, 1/2].$$

Its terminal area is $2F^2b^3/(3m) + Fb^2D/m$. To obtain the added area, shear the momentum segment vertically: each convex section lengthens by D over a transverse width Fb^2/m. As $b \to 0$, this area vanishes uniformly, but the central-record momentum width tends to $2(\sqrt{2}-1)FT$. Area closure alone therefore does not establish momentum recovery.

C072–C073: proof and assumptions, B39 audit. The finite-precision extension follows.

14. Finite precision and residual delay

A record $|q - y| \le \varepsilon$ clips the cut lens to a convex set C with position interval $[l, r]$. Let its increasing momentum-fibre endpoints be $L(q), U(q)$, its area A_C, its width $w = r - l$, and $D_* = U(r) - L(l)$. The exact terminal set is $S_b C + K_b$, with area

$$|S_b C + K_b| = A_C + \frac{2F^2b^3}{3m} + 2Fbw + \frac{Fb^2}{m}D_*.$$

After the inverse shear, each future-force segment adds a transverse width $w + sD_*/m$; integrating its extrusion gives the two mixed terms. Exact position and unobserved full-lens records recover the preceding formulas. Since $w \le 2\varepsilon$, $D_* \le 2Fa$ and $A_C \le 2Faw$,

$$|S_b C + K_b| \le 4FT\varepsilon + \frac{2F^2ab^2}{m} + \frac{2F^2b^3}{3m} \longrightarrow 0$$

for every joint precision/delay limit at fixed $F, m, T = a + b$. Central records still leave finite momentum uncertainty. R10 therefore tests two position records, separating recovery of both coordinates from area closure. C074–C075: complete formula and proof, B40 audit.

15. Two records and recovery of momentum

Positions recorded at $t_1 = T - b - \delta$ and $t_2 = T - b$, with errors bounded by $\varepsilon_1, \varepsilon_2$, give the momentum estimate $\widehat{p} = m(y_2 - y_1)/\delta$. Define

$$B = \frac{m(\varepsilon_1 + \varepsilon_2)}{\delta} + \frac{F\delta}{2}. \qquad R_p = B + Fb, \qquad R_q = \varepsilon_2 + \frac{bB}{m} + \frac{Fb^2}{2m}.$$

These errors of $\widehat{p}$ and $\widehat{q}_T = y_2 + b\widehat{p}/m$ are simultaneously attained by constant force and adverse record errors. The compatible-set area is at most $4R_qR_p$. If $\delta, b \to 0$ and $(\varepsilon_1+\varepsilon_2)/\delta \to 0$, both coordinates are recovered.

For equal errors, the established finite-difference optimum is $\delta_* = 2\sqrt{m\varepsilon/F}$. Choosing positive delay $b = \delta_*$ gives $R_p = 4\sqrt{mF\varepsilon}$, $R_q = 7\varepsilon$, and an action-error product $28\sqrt{mF}\varepsilon^{3/2} \to 0$. R11 tests minimax lower bounds. C076–C077: proof, exact prior-art match.

16. A gap that survives dense noisy records

At fixed position tolerance $\varepsilon > 0$, even the entire noisy position history leaves a sharp momentum uncertainty. Put $d = 2\sqrt{m\varepsilon/F}$, $t_2 = T - b$, $t_1 = t_2 - d$, and assume $t_2 \geq 2d$. Starting from zero displacement and velocity, forces $-F, +F$ for successive times $d/2$ prepare displacement $-\varepsilon$ with zero velocity at t_1. Force $+F$ for the next d reaches $+\varepsilon$ and momentum Fd. The entire observed displacement stays in $[-\varepsilon, \varepsilon]$.

This trajectory and its opposite have the same initial state and admit the same inertial position record, with opposite allowed record errors. Continuing their forces through the blind interval yields terminal half-separations

$$P_* = Fd + Fb, \qquad Q_* = \varepsilon + \frac{Fdb}{m} + \frac{Fb^2}{2m}.$$

Every estimator has worst-case coordinate errors at least P_*, Q_*. The preceding two-record estimator attains both; these are therefore exact coordinate minimax risks for two records and for the complete noisy history. At zero blind delay their product is $2\sqrt{mF}\varepsilon^{3/2} > 0$. Densifying records preserves this bound; improving their tolerance closes it.

The construction is established in Seeber–Haimovich Proposition 3.1. R11 connects it to reachable fibres and delayed phase prediction. The general principle is that, for symmetric convex inputs and bounded linear record errors, the central compatible fibre maximizes scalar output uncertainty: any compatible pair has an admissible half-difference, and the symmetric central pair realizes its diameter.

C078–C079: proof and precise information classes, B42 audit. The following composition test settles the proposed transfer to mass universality.

17. Worst-case composition and the cost of aggregate records

Product bounded-force/error classes add scalar minimax radii linearly. For positive masses with known initial states, complete position records of errors $\varepsilon_i > 0$, forces $|f_i| \leq F_i$ and common terminal time $T \geq \max_i 4\sqrt{m_i\varepsilon_i/F_i}$, put $M = \sum_i m_i$, $E = \sum_i m_i\varepsilon_i/M$ and $F_\Sigma = \sum_i F_i$. With all constituent records retained, the exact centre position and total momentum risks are

$$Q_A = E, \qquad P_A = 2\sum_i \sqrt{m_i F_i \varepsilon_i}.$$

The central compatible fibre is a Cartesian product. Its support in a scalar sum direction is the sum of the supports, attained by aligned symmetric extremizers; R11's midpoint argument converts that radius into exact minimax risk. For n identical constituents the product is therefore $H_A = nH_1$. It does not obey A08's invariant variance-coefficient law.

Keeping only the weighted sum of the records gives exactly the single-body force/error class (M, F_Σ, E). To lift any aggregate force f and error e back to the original constituents, choose $f_i = F_i f/F_\Sigma$ and $e_i = \varepsilon_i e/E$, preserving each initial state. Thus its exact risks are

$$Q_B = E, \qquad P_B = 2\sqrt{MF_\Sigma E} \geq P_A.$$

Equality holds precisely when $F_i/(m_i\varepsilon_i)$ is common to all constituents. Otherwise losing the individual records strictly increases the momentum risk in the same mechanical model.

If mass-only nonnegative coordinate radii are closed under product composition for every positive mass, additivity forces $Q(m) = q_*$, $P(m) = p_* m$. Their action product is extensive; a mass-independent product must be zero. Fixed acceleration bound $F(m) = am$ and common precision realize the positive extensive case, whose value still comes from a and the record band. R13 next examines the finite-horizon regime before the hidden pair can be prepared.

C080–C081: complete proof and information classes, B43 audit. These products multiply coordinate minimax risks and are not simultaneous error lower bounds on each motion.

18. Exact recovery before the preparation time

The complete noisy history has exact finite-horizon phase risks even before R11's hidden pair can be fully prepared. Keep known initial data, arbitrary measurable forces bounded by F, position errors bounded by epsilon, and no speed ceiling. With $\tau = \sqrt{m\varepsilon/F}$ and $s = T/\tau$,

$$Q(T) = \varepsilon \min\{s^2/2, 1\}, \qquad P(T) = \sqrt{mF\varepsilon}\, V(s),$$

$$V(s) = \begin{cases} s, & 0 \le s \le \sqrt{2}, \\ \sqrt{2s^2+4} - s, & \sqrt{2} \le s \le 4, \\ 2, & s \ge 4. \end{cases}$$

In scaled coordinates, the hidden path obeys $x(0) = x'(0) = 0$, $|x''| \le 1$, $|x| \le 1$. Its terminal velocity v is bounded by s. Putting negative acceleration first minimizes terminal position at fixed v, giving $x(s) \ge (v^2 + 2sv - s^2)/4$ and the middle upper bound. Integrating $x'(t) \ge v - (s-t)$ over the final v time units gives $v^2/2 \le 2$. The lower envelope of these three bounds is V. Constant positive acceleration attains the first branch. Negative acceleration for $h = s - \sqrt{s^2/2+1}$ followed by positive acceleration attains the middle; its minimum position is $-h^2 \ge -1$ and endpoint position is 1. The prepared R11 pair attains the final branch. Each momentum extremizer also maximizes endpoint position, so a blind interval b has exact risks

$$P_b = P(T) + Fb, \qquad Q_b = Q(T) + bP(T)/m + Fb^2/(2m).$$

Product constituent records still give $Q_A = \sum_i m_i Q_i / M$ and $P_A = \sum_i P_i$. Retaining only their centre record gives the single-body formula with (M, F_Σ, E) at every horizon. Position information is strictly lost exactly when some $F_i T^2/2 < m_i \varepsilon_i$ and another $F_j T^2/2 > m_j \varepsilon_j$. Thus the transient can lose both phase coordinates. For equal m and epsilon, forces F and 16F at $T = \sqrt{m\varepsilon/F}$ give

$$Q_A = 3\varepsilon/4, \quad Q_B = \varepsilon, \quad P_A = 9\sqrt{mF\varepsilon}, \quad P_B = (\sqrt{714} - 17)\sqrt{mF\varepsilon} > P_A.$$

Identical copies retain $H_A = H_B = nH_1(T)$. At fixed precision the early product is $F^2T^3/(2m)$; at fixed positive T it closes as precision improves. R14 tests a shared apparatus error budget that prevents freely aligned full-width errors. C082–C083: complete written proof and B44 bounded audit.

19. Shared error geometry controls composition

For n identical constituents with known initial phases, force ceiling F and complete position records, impose one pointwise apparatus budget $\|e(t)\|_r \le \varepsilon$, $1 \le r \le \infty$. Set $E_n = \varepsilon n^{-1/r}$, with $1/\infty = 0$. Averaging has error at most E_n by the norm inequality, and force at most nF at mass nm. Equal errors and forces lift every such scalar experiment. Equal displacement paths also lift its entire central compatible fibre. Thus retaining all records or only their average gives identical scalar centre minimax risks:

$$Q_n = E_n \min(s_n^2/2, 1), \qquad P_n = n\sqrt{mFE_n}\, V(s_n), \qquad s_n = T\sqrt{F/(mE_n)},$$

where V is section 18's function. For $s_n \ge 4$,

$$\boxed{H_n^{\mathrm{sat}} = n^{1-3/(2r)} H_1^{\mathrm{sat}}, \qquad H_1^{\mathrm{sat}} = 2\sqrt{mF}\,\varepsilon^{3/2}.}$$

A quadratic budget gives $n^{1/4}$; $r = 3/2$ gives copy-count invariance. R15 tests independent block budgets against this shared preparation. C084–C085: proof and scope, B45 audit.

20. Independent block apparatuses restore extensive scaling

For k independently budgeted blocks of a identical constituents, with the same pointwise ℓ^r error allowance epsilon per block, put $E = \varepsilon a^{-1/r}$ and $N = ka$. Product compatible fibres add their scalar target radii. Thus for all constituent records, all block-average records, or only the centre record, the risks equal the one-body risks at mass Nm, force bound NF and precision E. At $T \geq 4\sqrt{mE/F}$,

$$H_{\text{blocks}} = ka^{1-3/(2r)}H_1, \qquad H_{\text{global}} = (ka)^{1-3/(2r)}H_1.$$

Here the second apparatus has only one global allowance epsilon. At $r = 3/2$ the products are kH_1 and H_1 respectively. Matching global-budget centre risks with separate blocks requires reducing each allowance to $\varepsilon k^{-1/r}$. For finite r and $k > 1$ this still gives different error sets: a Cartesian product of block balls is strictly contained in the matching global ball. Equal centre risks do not establish equal preparations.

For unequal effective block precisions E_j, retained block records instead give

$$Q = \sum_j (n_j/N)q(T, E_j), \qquad P = \sum_j n_j p(T, E_j),$$

using section 18's one-constituent functions. Keeping only the centre record gives the risks at precision $\bar{E} = \sum_j (n_j/N)E_j$. In saturation, strict concavity of the square root makes momentum risk larger unless all E_j agree. R16 tests contraction of finite mechanical record errors at fixed apparatus parameters. C086–C087: proof and B46 audit.

21. Exact calibration at fixed nonzero coupling

Keep section 11's apparatus and one sufficiently small positive coupling λ fixed. Its nominal four-momentum map has the uniform expansion

$$F_\lambda(z) = -\lambda\mathcal{A}z + \mathcal{R}_\lambda(z), \qquad \|\mathcal{R}_\lambda\|_{C^1} \leq C\lambda^3.$$

Differentiating the smooth flow preserves the probe $O(\lambda)$ and receiver/clock reaction $O(\lambda^2)$ bounds. The integrated record remainder is therefore cubic in coupling also in first derivative. On a convex neighbourhood of the energy shell, the matrix margin alpha>0 gives

$$\|F_\lambda(z) - F_\lambda(w)\| \geq \tfrac{1}{2}\lambda\alpha\|z - w\|.$$

Let incoming apparatus uncertainty have width b, and the four final-record errors have width rho, in fixed component units. The exact nonlinear minimum-residual calibration on the compact receiver shell then satisfies

$$\|\hat{z} - z\| \leq \frac{4}{\lambda\alpha}(C_\lambda b + \rho).$$

Thus reconstruction error vanishes as b and rho shrink with coupling, masses and duration fixed. Momentum disturbance remains bounded by $CP_*(\lambda b + \lambda^2)$, so its product with position-reconstruction error also vanishes. Exact calibration and improving record access are supplied resources. R17 holds incoming width positive to test whether unknown probe momenta can hide distinct receiver states. C088–C089: proof and domain, B47 audit.

22. Positive preparation width leaves exact record ambiguity

Keep the four final momentum records as the only data, with a fixed positive incoming apparatus box of half-width b and sufficiently small coupling $\lambda > 0$. In fixed component units the derivative of final records with respect to incoming probe momenta u is uniformly within 1/2 of I. The nominal receiver derivative is bounded by $L\lambda$. Thus the map

$$T_w(u) = u - G_\lambda(w,u) + F_\lambda(z_*)$$

contracts the ball $\|u\| \le b/2$ into itself whenever $\|w - z_*\| \le b/(4L\lambda)$. Its fixed point supplies exact nonlinear record compensation, with an interior preparation margin.

At the shell point $z_* = (0, 0, \sqrt{2E/d}, 0)$, completing the square gives the local chart

$$y = \frac{g}{d}x + \sqrt{\frac{2E - (a - g^2/d)x^2 - P^2/\mu}{d}}, \qquad Q = 0.$$

Let r_0 be a fixed dimensionless chart-square radius and C_0 its Lipschitz bound in fixed length/momentum units L_*, P_*. With

$$r = \min(r_0, b/(4LC_0\lambda)),$$

the entire square $|x| \le L_* r$, $|P| \le P_* r$ has one common exact record. Every initial-state estimator therefore has worst-case errors

$$\epsilon_x \ge L_* r, \qquad \epsilon_P \ge P_* r, \qquad \mathcal{H}_{\mathrm{rec}} \ge L_* P_* r^2 > 0.$$

The compatible fibre's canonical (x, P) projection has area at least $4L_*P_*r^2$. This reconstruction bound depends on the positive product preparation support and four-record access; its small-b rate matches section 21's upper bound. R18 reveals incoming momenta and tests whether unknown probe positions still hide receiver changes. C090–C091: proof and domain, B48 audit.

23. Position uncertainty survives incoming momentum calibration

Revealing the four incoming probe momenta still leaves a common-record shell patch for one fixed admissible pulse design. On the zero-incoming-momentum, nominal-clock slice, the derivative with respect to initial probe positions is

$$D_q G_\lambda = \lambda^2 B(z) + O(\lambda^3).$$

Both receiver and clock reaction contribute to B. Ordered disjoint pulses make it lower triangular. For pulses $f_j(s_0 + vt) = \phi((t - t_j)/\delta)$ with $x_j = x^0(t_j) \ne 0$, its diagonal obeys

$$B_{jj} = -\frac{K^2 x_j^2 \delta}{M_c v^2} \int \phi^2 + O(\delta^2).$$

Choose independent signal times avoiding the isolated zeros of x^0 and then a sufficiently small positive duration δ, held fixed thereafter. Every diagonal is nonzero. The clock contribution, of order δ, dominates the receiver's order-δ^3 term.

In fixed component units, multiplication by $(\lambda^2 B(z_*))^{-1}$ gives a contraction on the position ball $b/2$ whenever $\|w - z_*\| \le \lambda b/(4C)$. Its fixed point preserves all eight initial and final momentum records exactly. Using section 22's shell chart gives

$$r = \min\left(r_0, \frac{\lambda b}{4CC_0}\right), \qquad \epsilon_x \epsilon_P \ge L_* P_* r^2, \qquad \mathrm{area}_{x,P} \ge 4L_* P_* r^2.$$

These bounds quantify the unknown position support; at fixed width their displayed lower bound closes with coupling. R19 adds final pointer positions and known initial clock position and momentum to test uniform joint recovery. C092–C093: proof and domain, B49 audit.

24. Full pointer records remove the fixed-position ambiguity

With initial clock data c and probe momenta u revealed, all eight final pointer coordinates determine both receiver state z and unknown incoming positions q. The initial positions occupy a fixed positive box. Clock data may range over a fixed sufficiently small known-data box, on which the free signal matrices A_c have a uniform inverse margin $\alpha > 0$. Free drift is $d_j(u) = Tu_j/M_j$.

In fixed component units the scaled exact record map satisfies

$$F(z,q) = \left(\frac{\pi(T) - u}{\lambda}, q(T) - d(u)\right) = (-A_c z, q) + R(z,q), \qquad ||R||_{C^1} \leq C\lambda.$$

The receiver and clock reactions are uniformly order λ on this fixed preparation domain. Integrating the pointer equations and their variations gives the displayed estimate on a convex neighbourhood of the energy ball times the position box. For $\beta = \min(\alpha, 1)$ and $C\lambda \leq \beta/2$, segment integration yields the global bound

$$||F(v) - F(w)|| \geq \tfrac{1}{2}\beta ||v - w||.$$

Thus exact joint records remove section 23's ambiguity. If final momentum and position errors are at most ρ_π and ρ_q in fixed units, a minimum-residual fit on the shell times the box gives

$$\epsilon_x \epsilon_P \leq \frac{16 L_* P_*}{\beta^2} \max(\rho_\pi/\lambda, \rho_q)^2 \longrightarrow 0$$

as both record errors vanish at fixed positive coupling. Preparation widths, masses and observation time stay fixed. Exact incoming clock/momentum data and improving simultaneous final position/momentum access are supplied resources. This is delayed reconstruction; disturbance need only stay bounded. R20 hides the initial clock data and tests an exact common-record shell family. C094–C095: proof and domain, B50 audit.

25. A hidden clock leaves an exact receiver phase interval

With clock offset and speed hidden, eight exact final pointer records admit an energy-preserving receiver phase family for a fixed admissible pulse design. Incoming probe momenta remain known at zero. Write clock data as $c = (s, v)$, receiver flow as $\exp(Mt)$, and $H_s(z) = z^T G z/2$. Section 24's scaled map extends smoothly to zero coupling. The implicit-function theorem solves

$$F_{\lambda,c}(z_\lambda(c), q_\lambda(c)) = Y_\lambda, \qquad z_0(c) = A_c^{-1} A_{c_0} z_*.$$

Interior supports give $\partial_s A = -AM/v$. Consequently the offset derivative is Mz_*/v_0 and preserves receiver energy at zero coupling. For four pulses at times of order a small design duration ε, the independent observation rows $e_x M^n$, $n = 0, 1, 2, 3$, give

$$\det A_{s_0,v} = D(r\varepsilon)^{10}(1 + O(\varepsilon)), \quad r = v_0/v, \quad D \neq 0,$$
$$\operatorname{tr}(-A^{-1}\partial_v A) = 10/v_0 + O(\varepsilon) > 0.$$

The exponent counts four integration factors and derivative orders $0+1+2+3$. Fix this design. Positive trace supplies a shell point with $\partial_v H_s(z_0(c)) > 0$ and both canonical phase derivatives nonzero. A second implicit equation therefore solves $H_s(z_\lambda(c)) = E$ for $v = v_\lambda(s)$, with

$$v'_\lambda(s_0) = O(\lambda), \qquad z'_\lambda(s_0) = Mz_*/v_0 + O(\lambda).$$

On a small fixed clock-offset interval of half-width σ, both canonical derivatives retain magnitudes at least $k_x, k_P > 0$. Clock-speed and incoming position adjustments are $O(\lambda\sigma)$, preserving preparation margins. Common-record endpoints then imply

$$\epsilon_x \epsilon_P \geq k_x k_P \sigma^2 > 0.$$

This action-valued product is a clock-preparation cost. The family is a curve, so the result asserts coordinate risks rather than positive canonical area. R21 reveals offset alone and tests local recovery through energy transversality. C096–C097: proof and domain, B51 audit.

26. Clock position and exact energy permit stable local recovery

Revealing initial clock position alone removes section 25's phase ambiguity on a fixed local receiver preparation patch. Initial clock speed remains unknown, incoming probe momenta are known at zero, and the initial receiver energy is supplied exactly. Use the same fixed transverse pulse design and shell point z_* as in section 25. In fixed component units, append energy to the eight scaled records:

$$\mathcal{G}_\lambda(z, q, v) = (F_\lambda(z, q, v), H_s(z)/E_*), \qquad F_0(z, q, v) = (-A_{s_0,v} z, q).$$

The derivative kernel at zero coupling obeys

$$dq = 0, \qquad dz = Bz_* \, dv, \qquad B = -A_0^{-1}(\partial_v A)_0, \qquad (z_*^T G B z_*) \, dv = 0.$$

The positive transverse factor established in section 25 makes the reference derivative J invertible. Put $\gamma = 1/\|J^{-1}\|$. Continuity and uniform C^1 convergence give one fixed convex neighbourhood U and $\lambda_0 > 0$ with

$$\sup_U ||D\mathcal{G}_\lambda - J|| \leq \gamma/2, \qquad ||\mathcal{G}_\lambda(w) - \mathcal{G}_\lambda(w')|| \geq \tfrac{1}{2}\gamma ||w - w'||.$$

The second inequality follows by integrating along the segment inside U. Choose a compact shell patch times independent positive probe-position and clock-speed boxes contained in U. On this physical set the energy outputs agree, so the eight records alone determine all unknowns. For final momentum and position error bounds ρ_π, ρ_q, a minimum-residual fit gives

$$||\widehat{w} - w|| \leq \frac{4}{\gamma} \max(\rho_\pi/\lambda, \rho_q), \qquad \epsilon_x \epsilon_P \leq \frac{16 L_* P_*}{\gamma^2} \max(\rho_\pi/\lambda, \rho_q)^2.$$

The canonical product has action units and closes with record error at fixed positive coupling and fixed local preparation. Initial clock momentum is recovered as well. The zero-coupling extension is a proof device; physical recovery requires a nonzero signal. Exact energy and knowledge of the local patch are supplied information. R22 tests distinct-speed common records across the entire shell. C098–C099: proof and domain, B52 audit.

27. Full-shell recovery fails at two distinct clock speeds

The local preparation patch in section 26 carries genuine information. With the full receiver energy shell admitted, a fixed early-pulse design gives distinct-speed preparations with identical eight final pointer records. Clock offset and receiver energy remain known.

Let O have rows $e_x M^n$, $n = 0, 1, 2, 3$. Factoring the pulse moment matrix before inversion gives, for a small fixed design duration ε,

$$B_\varepsilon = -A_{v_0}^{-1} \partial_v A_{v_0} = \frac{1}{v_0} O^{-1} \operatorname{diag}(1, 2, 3, 4) O + O(\varepsilon).$$

For $Q = 0$, its limiting energy response is

$$v_0 z^T G B^{(0)} z = g(x - y)(9x - 5y) + \frac{2P^2}{\mu}.$$

Take $x = L > 0$, $P = p > 0$ sufficiently small and vary y from zero to $7L/5$, normalizing the entire path onto $H_s = E$. The endpoint energy responses have opposite signs and both canonical coordinates stay positive. Choose the design first and then a small fixed speed separation $d_v > 0$. Equality of scaled zero-coupling records at v_0 and $v_1 = v_0 + d_v$ gives

$$z' = A_{v_1}^{-1} A_{v_0} z = z + d_v B_\varepsilon z + O(d_v^2).$$

The energy mismatch changes sign along the path, while both coordinate separations are bounded below by positive multiples of d_v. Intermediate value supplies an actual equal-energy pair.

At positive coupling solve the exact eight-record equation at the second speed for receiver state and incoming positions. Its invertible leading derivative gives uniform $O(\lambda)$ continuation along the compact path. For sufficiently small coupling the endpoint signs, preparation margins and canonical separations persist. Another intermediate-value argument gives exact equal-energy, equal-record states, and consequently

$$\epsilon_x \epsilon_P \geq \frac{c_x c_P d_v^2}{16} > 0.$$

This action-valued risk depends on the allowed speed interval and full-shell preparation. R23 adds the persistent final clock momentum to test global recovery from physical clock information. C100–C101: proof and domain, B53 audit.

28. A final clock-momentum record restores global recovery

Add the persistent final clock momentum to the eight pointer records, with initial clock position still known. The nine records determine receiver state, incoming probe positions and initial clock speed on a whole bounded domain. Exact receiver energy and the local-patch prior are unnecessary.

In fixed component units, including clock momentum unit $M_c V_*$, the map is

$$\mathcal{F}_\lambda(z, q, v) = (\pi(T)/\lambda, q(T), p_s(T)/M_c) = L(z, q, v) + R_\lambda, \qquad L = (-A_v z, q, v), \quad \|R_\lambda\|_{C^1} \leq C\lambda.$$

The last block includes the exact reaction

$$p_s(T) = M_c v - \lambda K \sum_j \int_0^T f_j'(s(t)) x(t) q_j(t)\, dt.$$

Let α be a uniform lower bound for A_v, and $D = \sup \|(\partial_v A_v) z\|$ on the fixed convex preparation domain. The speed and position blocks bound their input differences directly; the receiver block then yields

$$\|L(w) - L(w')\| \geq \beta \|w - w'\|, \qquad \beta = \min(1, \alpha/(1 + D)).$$

Subtracting the remainder's Lipschitz bound gives an exact inverse margin $\beta/2$ whenever $C\lambda \leq \beta/2$. This is a global estimate for a nonlinear leading map, rather than a pointwise rank argument. For final record errors ρ_π, ρ_q, ρ_c, a compact minimum-residual fit gives

$$\epsilon_x \epsilon_P \leq \frac{16 L_* P_*}{\beta^2} \max(\rho_\pi/\lambda, \rho_q, \rho_c)^2.$$

The action-valued error product closes with record precision at fixed coupling and positive preparation widths. Clock back-reaction is jointly calibrated; the final momentum is not substituted as an exact initial momentum. Joint record access remains supplied. R24 adds final clock position and hides initial offset. C102–C103: proof and domain, B54 audit.

29. Full clock records replace its initial preparation data

Add final clock position at the known time T, and make its initial offset unknown. With the initial probe momenta still known at zero, all ten final apparatus coordinates recover receiver state, incoming probe positions and both initial clock coordinates on a bounded domain.

Use clock units S_*, V_* and $\theta = TV_*/S_*$. The leading dimensionless map and exact remainder are

$$L(z, q, s, v) = (-A_{s,v}z, q, s + \theta v, v), \qquad \mathcal{F}_\lambda = L + R_\lambda, \quad \|R_\lambda\|_{C^1} \leq C\lambda.$$

The extra reaction term in physical clock position is exactly

$$s(T) - s - vT = -\frac{\lambda K}{M_c} \sum_j \int_0^T (T - t) f_j'(s(t)) x(t) q_j(t)\, dt.$$

Let d be the leading output difference. Inverting the clock shear gives $|\Delta v| \leq d$ and $|\Delta s| \leq (1+\theta)d$. With uniform signal inverse margin α and bounds D_s, D_v for its parameter derivatives applied to receiver states, the receiver difference obeys

$$\alpha\|\Delta z\| \leq [1 + D_s(1 + \theta) + D_v]d.$$

These bounds give a global leading inverse margin

$$\beta = \min\{(1 + \theta)^{-1}, \alpha/[1 + D_s(1 + \theta) + D_v]\} > 0.$$

The remainder reduces it by at most $C\lambda$, giving margin $\beta/2$ at small positive coupling. For final pointer and clock errors set $\delta = \max(\rho_\pi/\lambda, \rho_q, \rho_s, \rho_c)$; compact fitting yields $\epsilon_x\epsilon_P \leq 16L_*P_*\delta^2/\beta^2$.

Sections 24–29 isolate the information trade: final clock phase replaces initial clock calibration, while exact incoming probe momenta remain shared preparation data. The comparison table collects all six cases. R25 makes the entire incoming apparatus state unknown and tests compensation even with full final records. C104–C105: proof and domain, B55 audit.

30. Unknown full apparatus preparation hides the receiver shell

With all incoming apparatus coordinates unknown in a fixed positive box, all ten exact final coordinates admit receiver-shell ambiguity. At sufficiently weak coupling, the worst-case coordinate risks equal their no-record values.

Let η denote the ten initial apparatus coordinates in fixed component units, centred at η_*. The unscaled final-record map G_λ has free limit $S\eta$, where S sends each apparatus pair (q, p) of mass M to $(q + Tp/M, p)$. Thus $F_\lambda = S^{-1}G_\lambda$ satisfies uniformly on the convex receiver energy ball and a small fixed apparatus box

$$\|D_\eta F_\lambda - I\| \leq 1/2, \qquad \|D_z F_\lambda\| \leq L\lambda.$$

For a nominal shell point z_*, the map $\eta \mapsto \eta - F_\lambda(w, \eta) + F_\lambda(z_*, \eta_*)$ contracts the radius-$b/2$ preparation ball whenever $\|w - z_*\| \leq b/(4L\lambda)$. Its fixed point gives

$$G_\lambda(w, \eta(w)) = G_\lambda(z_*, \eta_*), \qquad \|\eta(w) - \eta_*\| \leq 2L\lambda\|w - z_*\| \leq b/2.$$

At fixed $b > 0$, take $\lambda \leq \min(\lambda_0, b/(4LD))$, where $D = \max_{H_s(w)=E} \|w - z_*\|$. One full record then hides the whole shell. Writing $k_x = a - g^2/d > 0$, its canonical projection is the ellipse $k_x x^2 + P^2/\mu \leq 2E$. Common-record endpoints force both semiaxis errors; the constant estimator $(0, 0)$ attains them over the entire admitted class:

$$R_x = \sqrt{2E/k_x}, \qquad R_P = \sqrt{2\mu E}, \qquad \inf_{(\widehat{x},\widehat{P})} \epsilon_x \epsilon_P = 2E\sqrt{\mu/k_x}.$$

The projected compatible area is π times this action-valued product. The fixed-width weak-coupling condition supplies the admissibility margin; receiver energy and stiffness supply the saturated value. Exact initial apparatus energy is not supplied. R26 imposes that extra scalar constraint on $\eta(w)$. C106–C107: proof and limits, B56 audit.

31. Exact initial energies leave an antipodal ambiguity

Supply the initial apparatus energy $H_0 = M_c v_0^2/2$ as well as receiver energy E. An exact common-record receiver pair still exists. Choose a receiver shell point z_* and prescribe terminal zero probe phase, clock position $s_0 + v_0 T$, and

$$z(T) = r\Phi_T z_*, \qquad p_s(T) = \sqrt{2M_c[H_0 + E(1-r^2)]}.$$

Terminal total energy is exactly $E + H_0$. Backward Hamiltonian flow gives initial receiver state $Z_\lambda(r)$; at zero coupling its energy is $r^2 E$. The derivative $2E$ at $r = 1$ supplies an implicit adjustment with $H_s(Z_\lambda(r_\lambda)) = E$. Endpoint pulses vanish, so conservation gives initial apparatus energy H_0 exactly. Terminal zero probes imply

$$r_\lambda = 1 + O(\lambda^2), \qquad Z_\lambda = z_* + O(\lambda^2),$$

while the full incoming apparatus displacement is $O(\lambda)$ and fits inside half the fixed preparation box for sufficiently small coupling.

The simultaneous sign involution $(z, q, \pi, s, p_s) \mapsto (-z, -q, -\pi, s, p_s)$ preserves the equations in the linear cutoff regions. It leaves the clock and both quadratic energies unchanged. Because terminal probes vanish, both preparations have identical complete final apparatus records and exactly opposite initial receiver states.

Put $k_x = a - g^2/d$ and choose $z_* = (\sqrt{E/k_x}, \sqrt{\mu E}, (g/d)\sqrt{E/k_x}, 0)$. Common-record endpoints then give

$$\epsilon_x \epsilon_P \geq |(Z_\lambda)_x (Z_\lambda)_P| \longrightarrow E\sqrt{\mu/k_x} > 0.$$

The limit concerns the displayed lower bound, with action units. It is energy-dependent and supplies neither exact minimax saturation nor positive area. R27 tests a known nonzero initial probe displacement; breaking this involution must be distinguished from proving global recovery. C108–C109: proof and domain, B57 audit.

32. One calibrated displacement leaves full receiver ambiguity

Fix a known nonzero incoming probe displacement $q_1 = c$, with $|c|/L_1 < b/4$, while retaining exact initial energies E, H_0 and all ten final apparatus coordinates. Full receiver recovery still fails for sufficiently small positive coupling. The new proof uses a preparation chart rather than the sign involution, which would send c to $-c$.

Before the pulses, solve the apparatus energy for positive clock momentum,

$$p_s = \sqrt{2M_c\left(H_0 - \sum_{j=1}^{4} \frac{\pi_j^2}{2M_j}\right)}.$$

Eight independent apparatus coordinates remain: q_2, q_3, q_4, the four probe momenta and s. On the receiver shell use x, P, Q near zero and solve

$$y = \frac{g}{d}x + \sqrt{\frac{2E - k_x x^2 - P^2/\mu - Q^2/\nu}{d}}, \qquad k_x = a - g^2/d > 0.$$

In fixed component units this gives an injective eleven-coordinate chart Ψ containing a closed parameter ball of radius $r > 0$, independent of coupling, with strict radicands and half-box preparation margins. Its boundary is S^{10}. The continuous map taking each chart point to its ten final apparatus coordinates has an equal-image antipodal pair by Borsuk–Ulam. Both preparations have exactly the supplied energies and calibration.

For the inverse-shear record F_λ, the uniform ambient bounds give

$$\|D_\eta F_\lambda - I\| \le \tfrac{1}{2}, \qquad \|D_z F_\lambda\| \le L\lambda.$$

Hence equal records imply $\|\eta_+ - \eta_-\|_2 \le 2L\lambda\|z_+ - z_-\|_2$. The eleven chart coordinates are a projection of the full initial state; the antipodal parameter distance is $2r$. Therefore the full receiver-state minimax risk in these fixed Euclidean component units obeys

$$R_z \ge \frac{1}{2}\|z_+ - z_-\|_2 \ge \frac{r}{\sqrt{1 + 4L^2\lambda^2}} > 0.$$

This is a dimensionless full-state error bound. The difference may lie in internal coordinates; R28 tests separation in both x and P and a possible canonical error product. C110–C111: proof and margins, B58 theorem audit.

33. Calibrated records still hide both canonical coordinates

A fixed smooth pulse design preserves a positive canonical risk product under R27's nonzero calibration $q_1 = c$, both exact initial energies and all ten final apparatus coordinates. Let η_c have only $q_1 = c$ nonzero among probe coordinates. The exact record compensator about a receiver shell point $\bar{z}$ satisfies

$$G_\lambda(w, \eta_\lambda(w)) = G_\lambda(\bar{z}, \eta_c), \qquad \|\eta_\lambda(w) - \eta_c\| \le 2L\lambda\|w - \bar{z}\|.$$

Put $h(t) = f_1(s_0 + v_0 t)$. Dividing the calibration and initial apparatus-energy residuals by λ gives smooth zero-coupling limits $-(K/M_1)A(w - \bar{z})$ and $-KcB(w - \bar{z})$, where

$$A(w) = \int_0^T t h(t) x_w(t)\, dt, \qquad B(w) = \int_0^T h(t)\dot{x}_w(t)\, dt.$$

A first pulse near a small fixed τ makes the normalized rows approximate $x(\tau)$ and $P(\tau)/\mu$. Their kernel contains a direction close to $\Phi_{-\tau}(0, 0, Y, 0)$, whose initial canonical components are $gY\tau^2/(2\mu) + O(\tau^4)$ and $-gY\tau + O(\tau^3)$. Both are nonzero. A positive fixed pulse width preserves this property; three further disjoint pulses retain R06's four-record signal rank.

In the two-dimensional kernel plane choose $\bar{z}$ on $H_s = E$ orthogonal to that direction v in the energy inner product. Then dH_s, A, B are independent. The implicit-function theorem applied to the divided residuals, receiver energy and a curve parameter gives exact common-record states $w_\lambda(t)$. A fixed rectangle $|t| \le \delta$, $0 < \lambda \le \lambda_1$ retains preparation margins and canonical derivatives of magnitude at least $|v_x|/2$ and $|v_P|/2$. Its endpoints imply

$$\epsilon_x \epsilon_P \ge \frac{\delta^2 |v_x v_P|}{4} > 0.$$

The action-valued bound is uniform on this coupling interval and depends on preparation, energy and pulse design. C112–C113: proof and limits, B59 audit. R29 adds a second calibrated displacement and tests local versus global recovery.

34. A second calibration leaves two locally recoverable branches

Add $q_2 = 0$ while retaining $q_1 = c \neq 0$, both exact initial energies and the ten final apparatus records. The new divided compensator residual has limit $-(K/M_2)A_2(w - z_{\text{bar}})$, where $A_2 = \int t f_2 x_w$. The three linear rows are now A_1, B_1, A_2.

Choose narrow positive pulses near τ and 2τ. Their limiting rows are $x(\tau), \dot{x}(\tau), x(2\tau)$. On their kernel the state at τ has form $(0, 0, Y, Q)$, with $Y = -Q\tau/(3\nu) + O(\tau^3)$, and its initial canonical components obey

$$v_x = -\frac{gQ}{3\mu\nu}\tau^3 + O(\tau^5), \qquad v_P = \frac{5gQ}{6\nu}\tau^2 + O(\tau^4).$$

Fix sufficiently small times and positive widths, preserving rank three and these nonzero components. The resulting kernel line meets the receiver energy shell at $\pm z_{\text{bar}}$. At both points the derivative rows dH_s, A_1, B_1, A_2 are independent. The exact scaled residual equations therefore have two continued roots

$$w_+(\lambda) = z_{\text{bar}}, \qquad w_-(\lambda) = -z_{\text{bar}} + O(\lambda).$$

The compensating apparatus preparations preserve both calibrations, both energies, one exact complete record and the positive box margins. At each fixed positive coupling, eliminating the apparatus-record derivative proves a local inverse of the augmented map around each root. It does not choose between them globally. Common-record endpoint inequalities give

$$\epsilon_x\epsilon_P \geq \tfrac{1}{4}|(w_+ - w_-)_x(w_+ - w_-)_P| \longrightarrow |(z_{\text{bar}})_x(z_{\text{bar}})_P| > 0.$$

This is a preparation-dependent action-risk lower bound, not an area or exact minimax value. R30 adds $q_3 = 0$ and tests a uniform global inverse, including its record conditioning. C114–C115: proof and domain, B60 audit.

35. Three calibrations restore global recovery on a fixed box

Calibrate $q_1 = c \neq 0$, $q_2 = q_3 = 0$ and supply exact initial apparatus energy H_0, together with all ten final apparatus coordinates at known T. On a bounded convex receiver energy ball, a sufficiently small fixed box about the calibrated apparatus centre admits a uniform global inverse. The exact receiver energy is unnecessary. The box may be smaller than R29's original preparation box; its radius stays fixed in the precision limit.

Choose a third positive smooth pulse detecting the kernel of A_1, B_1, A_2. Analytic observability guarantees such a pulse, so the four rows

$$L = (-KA_1/M_1, -KA_2/M_2, -KA_3/M_3, -KcB_1)^T$$

are independent. Write $a = S^{-1}Y$ for the inverse free shear of a final record, and let $\eta_\lambda(w, a)$ be the uniform apparatus inverse chart. For $C = (q_1, q_2, q_3, H_{\text{app}})$ the map

$$D_\lambda(w, a) = \frac{C(\eta_\lambda(w, a)) - C(a)}{\lambda}$$

extends jointly smoothly to zero coupling. Its receiver derivative at the nominal record is L throughout the receiver ball. Shrinking the apparatus neighbourhood and coupling gives a uniform margin $\|L^{-1}(D_w D_\lambda - L)\| \leq 1/2$. Segment integration over the convex receiver domain then proves global injectivity for the exact constrained class.

For two admitted records the same estimate gives $\|w - w'\| \leq K_z\|Y - Y'\|/\lambda$. Minimum-residual fitting of a record with error at most δ therefore yields, in physical units,

$$\epsilon_x\epsilon_P \leq 4L_*P_*K_z^2(\delta/\lambda)^2.$$

This action-unit product closes at fixed positive coupling as record error vanishes, with preparation width fixed. Joint limits require $\delta/\lambda \to 0$ for this bound. C116–C117: proof and domain, B61 audit. R31 tests errors in the supplied calibrations and apparatus energy; the current theorem assumes both exact.

36. Calibration tolerance sets the recovery crossover

On section 35's unchanged apparatus box and receiver ball $Z = \{H_s \le E_{\max}\}$, allow deterministic errors ϵ in the four supplied quantities $C = (q_1, q_2, q_3, H_{\mathrm{app}})$ and δ in all ten final records. All component units, nonzero nominal displacement, pulse widths and masses remain fixed. Comparing the actual C values in the divided inverse chart gives

$$\|w - w'\| \le \frac{K}{\lambda}(\|C - C'\| + \|Y - Y'\|).$$

A compact feasible-set estimator therefore has

$$\|\widehat{w} - w\| \le \frac{2K}{\lambda}(\epsilon + \delta), \qquad \epsilon_x \epsilon_P \le 4 L_* P_* K^2 \left(\frac{\epsilon + \delta}{\lambda}\right)^2.$$

An exact common-record construction supplies a matching lower rate for $\delta = 0$. Take $Y_\lambda = G_\lambda(0, \eta_c)$ and reported calibration $C_* = C(\eta_c)$. The apparatus inverse chart obeys

$$G_\lambda(w, \eta_\lambda(w)) = Y_\lambda, \qquad \|\eta_\lambda(w) - \eta_c\| \le B\lambda\|w\|.$$

A fixed coupling ceiling keeps this compensator inside half the preparation box for every $w \in Z$. Let $M > 0$ bound $\|DC\|$ and $R_Z = \max_Z \|w\|$. Then the same reported data admit every receiver state in tZ, where

$$t = \min\{1, \epsilon/(MB\lambda R_Z)\}.$$

Writing $k_x = a - g^2/d > 0$, the canonical extrema of Z are $X_E = \sqrt{2E_{\max}/k_x}$ and $P_E = \sqrt{2\mu E_{\max}}$. Common-record endpoints force every estimator to satisfy $\epsilon_x \ge tX_E$ and $\epsilon_P \ge tP_E$. Together with feasible estimates and coordinatewise zero estimates, this proves that the optimal canonical product has order $\min\{1, (\epsilon/\lambda)^2\}$, up to fixed positive action-unit constants. When $\epsilon \ge MB\lambda R_Z$, the zero estimator attains the lower bounds:

$$R_x = X_E, \qquad R_P = P_E, \qquad \inf \epsilon_x \epsilon_P = 2E_{\max}\sqrt{\mu/k_x}.$$

This is a sufficient saturation threshold. The projected compatible area is $\pi X_E P_E$; the risk product has no 2π normalization. All four supplied constraints have positive tolerance in the lower construction. R32 retains three exact constraints and tests the canonical fibre left by one uncertain quantity. The scale here depends on receiver energy and mechanical parameters; at fixed coupling calibration refinement closes it. C118–C119: proof and domain, B62 information-radius audit.

37. One uncertain calibration: a curve instead of a ball

Keeping three supplied constraints exact leaves a common-record curve when the fourth has positive tolerance. For the exact coupled reference through receiver zero and nominal apparatus, let

$$T_\lambda(w) = \frac{C(\eta_\lambda(w, a_\lambda)) - C_*}{\lambda}.$$

Its smooth extension satisfies $T_\lambda(0) = 0$ and $T_0(w) = Lw$. The uniform local inverse supplies fixed $s_0 > 0$ and an exact family

$$w_{\lambda,j}(s) = T_\lambda^{-1}(se_j), \qquad |s| \le \min\{s_0, \epsilon/\lambda\}.$$

All ten final apparatus coordinates agree; only supplied constraint j changes, by λs. The leading receiver direction is $d_j = L^{-1}e_j$. For apparatus-energy uncertainty, $j = 4$, the three position-response rows annihilate this vector. Its two canonical entries decide the endpoint test. If both are nonzero, shrinking the common local interval gives

$$\epsilon_x \epsilon_P \ge \frac{L_* P_*}{4} |(d_j)_x (d_j)_P| \min\{s_0, \epsilon/\lambda\}^2.$$

Together with section 36's upper bound, this gives matching product order conditional on the two-entry test. The smooth projected curve has zero planar area even when both coordinate diameters are positive. A positive estimation product therefore need not be the area of the compatible states. Section 38 supplies a pulse selection satisfying all four nonvanishing tests. C120: full derivation, B63 audit.

38. Early pulses give four canonical ambiguity directions

Choose the first three pulse centres at $\tau, 2\tau, 3\tau$ for a sufficiently small fixed $\tau > 0$, then fix sufficiently narrow positive smooth widths. This explicit subfamily of section 35 makes both canonical entries of every $L^{-1}e_j$ nonzero. Section 37 therefore gives matching risk-product order $\min\{1, (\epsilon/\lambda)^2\}$ for any one uncertain supplied constraint, including apparatus energy, with the other three exact.

For free output x, put $X(u) = x(\tau u)$. The receiver equations give $X^{(4)} + \alpha\tau^2 X'' + \beta\tau^4 X = 0$, where $\alpha = a/\mu + d/\nu$ and $\beta = (ad - g^2)/(\mu\nu)$. In scaled initial-jet coordinates, the map $(x(\tau), x(2\tau), x(3\tau), \tau\dot{x}(\tau))$ is an $O(\tau^2)$ perturbation of cubic Hermite interpolation at $1, 2, 3$ and derivative at 1. Its cardinal polynomials have initial value/derivative pairs

$$(-3/2, 23/4), \quad (3, -7), \quad (-1/2, 5/4), \quad (-3, 11/2).$$

All are nonzero. Momentum is μ/τ times the second entry. After dividing pulse rows by their nonzero integral/time factors, supports of half-width at most h perturb this matrix by at most $C_\tau h$. The inverse identity bounds the inverse perturbation by $2\|\mathcal{H}_\tau^{-1}\|^2 C_\tau h$ when $\|\mathcal{H}_\tau^{-1}\| C_\tau h < 1/2$. Choose h to preserve the eight entry margins and both required signal determinants. Analytic observability selects the fourth disjoint pulse completing the ordinary signal matrix. All widths and the resulting small apparatus box stay fixed thereafter.

For an arbitrary R30 design the criterion is nonvanishing of $\mathrm{cof}_{j,1}(L)$ and $\mathrm{cof}_{j,2}(L)$. The selected family proves simultaneous feasibility; the action-unit constants remain design- and preparation-dependent. C121: full polynomial and pulse proof, B64 audit.

39. Correlated errors suppress the canonical fibre

For the correlated report $C - C_* = \alpha L e_y$, $|\alpha| \le \epsilon$, the exact reference-record curve has a hidden-coordinate limit:

$$w_\lambda(s) = T_\lambda^{-1}(sLe_y), \qquad w_0(s) = se_y, \qquad |s| \le s_\epsilon = \min(s_0, \epsilon/\lambda).$$

Here the pulses, box, scalar error normalization and local interval are fixed. Smooth mixed derivatives and $w_\lambda(0) = 0$ give $\|w_\lambda(s) - se_y\| \le K_1\lambda|s|$. The optimal scalar errors conditional on this compact common-record segment consequently satisfy

$$R_x^{\mathrm{loc}} R_P^{\mathrm{loc}} \le L_* P_* K_1^2 \min(\lambda s_0, \epsilon)^2.$$

Hidden-state ambiguity persists at fixed positive tolerance, while these canonical ranges close with coupling. This conditional estimate concerns one local fibre, rather than a global estimator across all final records.

To identify the first surviving term, retain the moving coupled reference in $N(\lambda, w) = C(\eta_\lambda(w, a_\lambda)) - C_*$ and put $V(w) = \frac{1}{2}\partial_\lambda^2 N(0, w)$. The inverse expansion is

$$w_\lambda(s) = se_y - \lambda L^{-1} V(se_y) + O(\lambda^2 |s|).$$

The mechanical tangent coefficient vanishes exactly. On the fixed linear-cutoff chart, receiver reversal with coupling reversal preserves every apparatus record. The coupled reference is even in coupling, so the unique compensator gives $N(-\lambda, -w) = N(\lambda, w)$ and $DV(0) = 0$. Backward terminal variational equations, after subtracting the moving coupled reference, give a homogeneous quadratic map V; explicit finite-width pulse integrals appear in the linked derivation, equations (10)–(13).

Put $\beta = -\Pi_{x,P} L^{-1} V(e_y)$ and $S = \min(s_0, \epsilon/\lambda)$. Then

$$\Pi_{x,P} w_\lambda(s) = \lambda s^2 \beta + O(\lambda^2 |s|), \qquad R_x^{\mathrm{loc}} R_P^{\mathrm{loc}} \le L_* P_* K_2^2 (\lambda S^2 + \lambda^2 S)^2.$$

Both beta components remain to be tested for the selected pulses. If both are nonzero and S/lambda is sufficiently large, centre-to-endpoint comparison gives a lower bound $L_* P_* |\beta_x \beta_P| \lambda^2 S^4 / 16$. Opposite endpoints cancel the displayed curvature term. These are conditional local-fibre statements. C122–C123: derivation and units, B65 audit, B66 audit.